

Gas Field Policy 17: Low Temperature Service

GENERAL:

High Speed Reciprocating Compressors with standard construction may be applied to temperatures -20°F and above.

Applications below -20°F must be reviewed by Siemens Energy Marketing Group prior to quoting. (In most cases, aluminum pistons must have a slightly oversize O.D.; the allowable cylinder working pressure and differential pressure limits must be reduced; an ethylene-glycol static cylinder jacket cooling must be used with all water cooled cylinders.

INLET GAS PREHEAT:

A: Cylinders with Gas Cooling:

The following method for estimating inlet preheat applies to all Siemens Energy gas field compressors using cylinders with gas cooling.

$$T'_i = T_i + \left[\left(R_c^{\frac{K-1}{K}} - 1 \right) \frac{(T_i + 459.59)}{20} \right] + \frac{(T_a - T_i)}{20}$$

Where:

$K = C_p/C_v$

$T_i =$ Given Inlet Temperature (°F)

$T_a =$ Ambient Temperature (°F)

$R_c =$ Ratio Of Compression

$T'_i =$ Preheated Inlet Temperature (°F)
(Use T'_i As Inlet Temperature For Performance Calculations)

B: Cylinders with forced Cooling:

The following method for estimating inlet preheat applies to all Siemens Energy gas field compressors using cylinders with forced cooling.

$$T'_i = T_i + \left[\left(R^{\frac{K-1}{K}} - 1 \right) \frac{(T_i + 459.59)}{20} \right] + \frac{(T_w - T_i)}{20}$$

Where:

$K = C_p/C_v$

$T_i =$ Given Inlet Temperature (°F)

$T_w =$ Incoming Jacket Water Temperature (°F)

$R_c =$ Ratio Of Compression

$T'_i =$ Preheated Inlet Temperature (°F)
(Use T'_i As Inlet Temperature For Performance Calculations)